How-To Library

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Creating a Custom Logger to Log Database Access Outside of Business Hours

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Abstract

This article describes how to create a custom logger and appender to log database access that occurs outside of normal business hours.

Supported Versions

- Dynamic Data Masking 9.6.0 or later

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Overview

You can create custom loggers and appenders in the Management Console to log security rule events.

Your organization has business hours from 8:00 AM to 5:00 PM. You want to audit database access that occurs outside of business hours.

In this example, you perform the following tasks:

1. Create a custom logger in the Management Console.
2. Create a custom appender.
3. Create a connection rule that directs requests to a security rule set.
4. Create a security rule set.
5. Create a security rule that logs database access based on the time of day.
6. View the log files.

Step 1. Create the Logger

In the Management Console, create a custom logger in the Management Console.

1. In the Management Console tree, select the Loggers node and click **Tree** > **Add Logger**.

   The **Add Logger** window appears.

   Note: Only a Dynamic Data Masking administrator can create or edit logger nodes. An administrator cannot delegate privileges to a non-administrator user.

2. Enter the name of the logger. The logger name can contain alphabetic characters, numbers, and underscores (_), and must not exceed 60 characters. In this example, the logger name is DatabaseAccess.

   Note: Remember the name of the logger that you enter. You use the logger name when you create the security rule.

3. Click **OK** to close the **Add Logger** window.

   The logger appears in the Management Console tree.

Step 2. Create the Appender

Add an appender to the logger. The appender defines how you want Dynamic Data Masking to save log events.

1. In the Management Console tree, select the logger that you created and click **Tree** > **Add Appender**.

   The **Add Appender** window appears.

2. Configure the following appender parameters:

   Type

   Select **Rolling File**. The rolling file appender type saves the log information to a text file.

   Name

   Enter the name of the appender. The appender name can contain alphabetic characters, numbers, and underscores (_), and must not exceed 60 characters. In this example, the logger name is AccessLog.

   File

   Enter the file path and file name of the log file. You can use any plain text extension, such as .log or .txt. If you do not specify a complete file path, the path originates in the Dynamic Data Masking installation directory. In this example, the file is `log/AccessLog.log`.

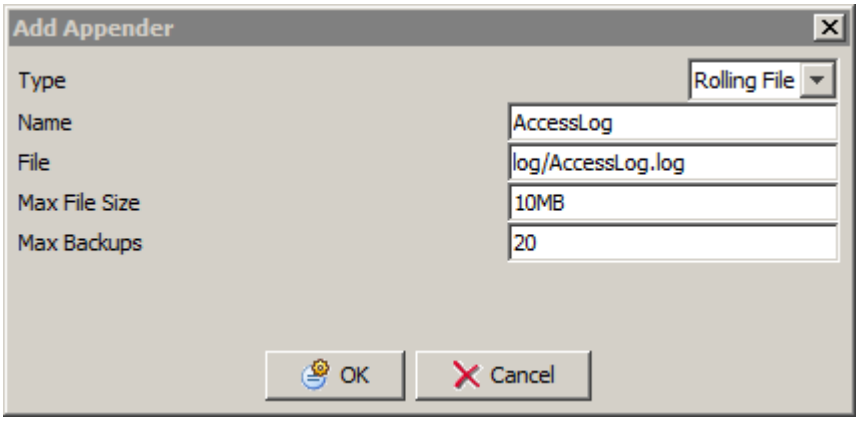
   Max File Size

   Enter the maximum size that you want the output file to reach before the appender creates a new file. In this example, the maximum file size is the default, 10MB.

   Max Backup

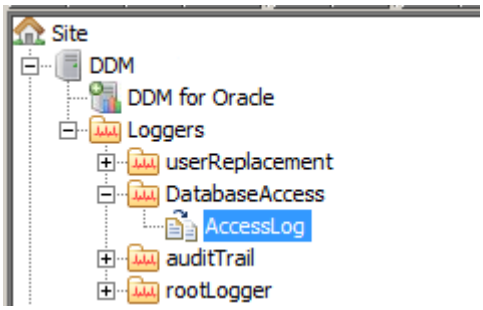
   Enter the number of backup files that the appender creates before it overwrites the oldest file. In this example, the maximum backup number is the default, 20.

   The following image shows the appender parameters:



3. Click **OK**.

   The appender appears in the Management Console tree. The following image shows the Management Console tree with the custom logger and appender:



Step 3. Create the Connection Rule

Create a connection rule that directs incoming database requests to the security rule set.

Note: You must have a Dynamic Data Masking database service added in the Management Console and a database connection created before you create the connection rule.

1. In the Management Console, select the Dynamic Data Masking service for the database that you want to log access to, and click **Tree** > **Connection Rules**.

   The **Rule Editor** window appears.

2. Click **Action** > **Append Rule**.

   The **Append Rule** window appears.

3. In the **Append Rule** window, configure the following parameters:

   Rule Name

   Enter the name of the rule. In this example, the rule name is UseRuleSet.

   Matcher

   Select **All Incoming Connections**. The All Incoming Connections matcher applies the connection rule action to all incoming database requests.

   Action

   Select **Use Rule Set**. The Use Rule Set action directs the requests to the security rule set that you define in the **Rule Set Name** field.

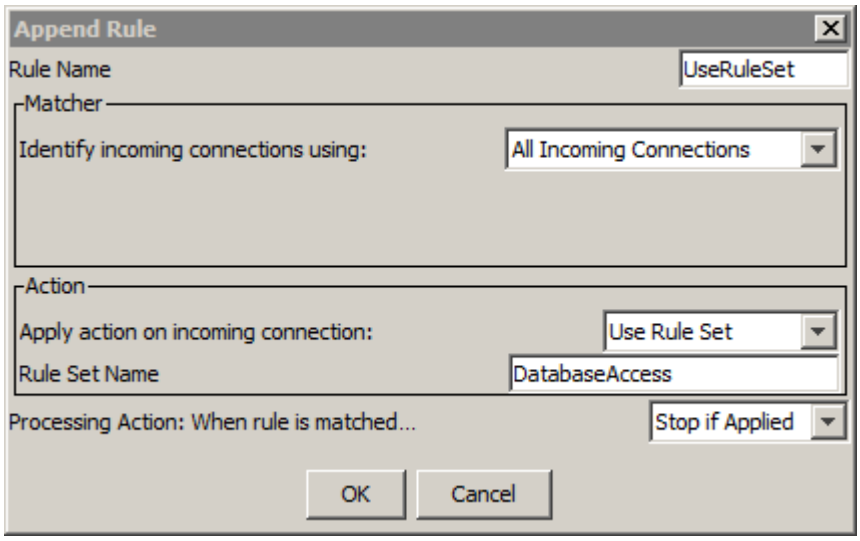
   Rule Set Name

   Enter the name of the security rule set that you want to create to log access to the database. In this example, the rule set name is DatabaseAccess.

   Processing Action

   Select **Stop if Applied**. The Stop if Applied processing action instructs the Rule Engine not to continue to the next rule in the tree.

   The following image shows the connection rule parameters:



4. Note the **Rule Set Name** value that you entered for the connection rule. You use the value to name the security rule set.
5. Click **OK** to close the window.
6. Click **File** > **Update Rules** to save the connection rule and click **File** > **Exit** to close the Rule Editor.

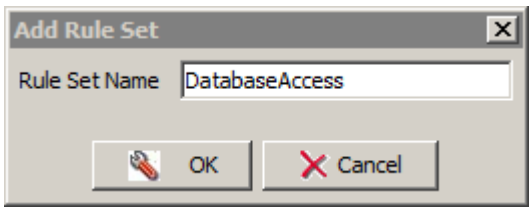
Step 4. Create the Security Rule Set

Create the security rule set to hold the security rule.

1. In the Management Console, select the root domain node and click **Tree** > **Add Rule Set**.

   The **Add Rule Set** window appears.
2. In the **Rule Set Name** field, enter the name of the rule set. Enter the name that you entered in the connection rule **Rule Set Name** filed. In this example, the rule set name is DatabaseAccess.

   The following image shows the **Add Rule Set** window:



3. Click **OK**.

   The rule set appears in the Management Console tree.

Step 5. Create the Security Rule

Create a security rule that logs database access outside of business hours.

1. In the Management Console tree, select the security rule set that you created and click **Tree** > **Security Rule Set**.

   The **Rule Editor** window appears.

2. In the Rule Editor, click **Action** > **Append Rule**.

   The **Append Rule** window appears.

3. Configure the following parameters for the security rule:

   Rule Name

   Enter the rule name. In this example, the rule name is LoggerRule.

   Description

   Enter an optional description of the rule.

   Matcher

   Select **Time of Day**. The Time of Day matcher defines a time frame during which the Rule Engine matches requests. Configure the **From Time**, **To Time**, and **Time Zone** parameters to define the hours during which you want to log database access. In this example, the matcher matches the request from 17:00 to 08:00 in the America/Los_Angeles time zone.

   Action

   Select **Log Message**. The Log Message action uses the logger that you defined in the Management Console to log security rule events. Configure the following parameters for the Log Message action:

   - Logger Name. Enter the name of the logger that you created in the Management Console. In this example, the logger name is DatabaseAccess.
   - Send As. Select the log level for the security rule event. The log level of the security rule must be a greater severity or equal severity of the log level that the Dynamic Data Masking administrator defines on the Dynamic Data Masking Server node of the Management Console tree. If the log level of the security rule is lower, the Rule Engine does not log the security rule event. In this example, the Rule Engine sends the event log data as information.
   - Logger Message. Enter a message that appears in the log output. In this example, the logger message is "A database request occurred outside of business hours."

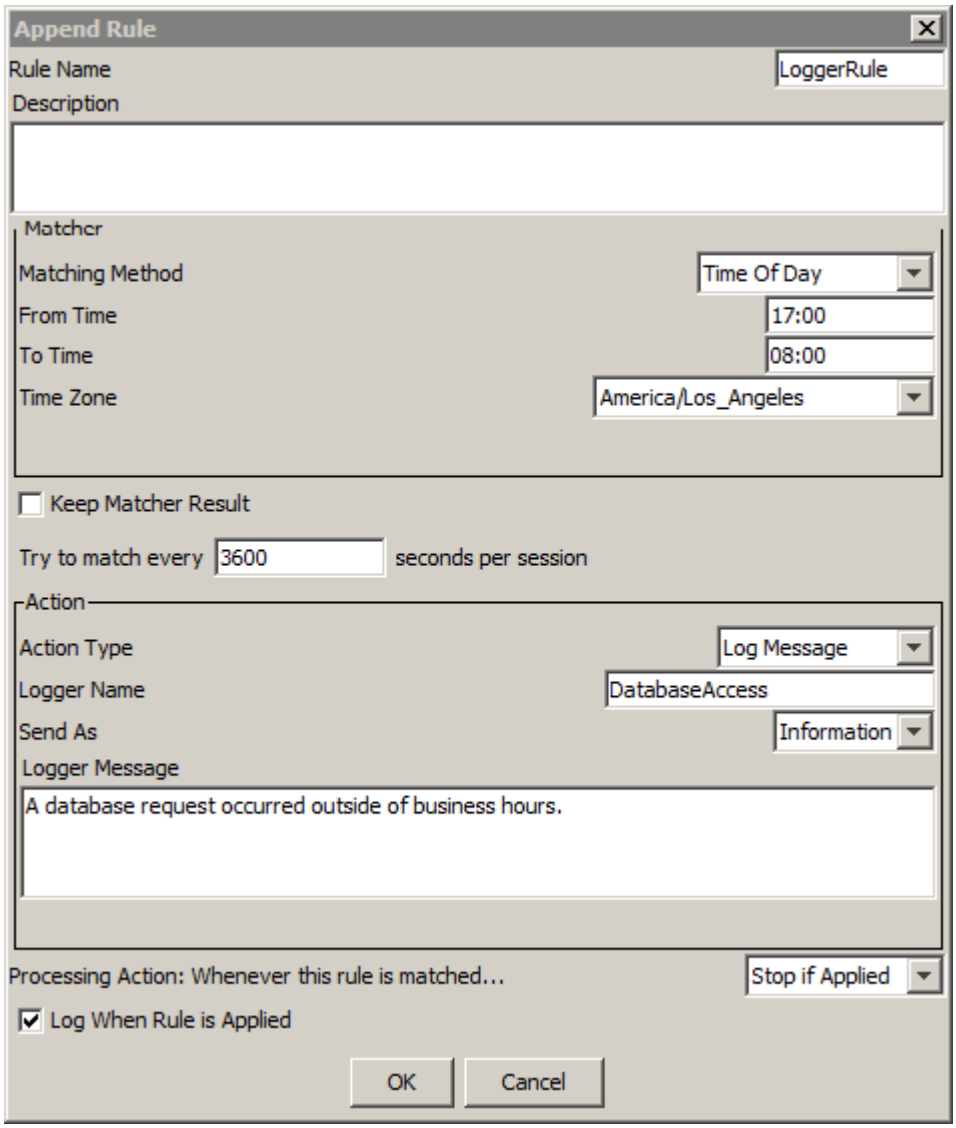
   Processing Action

   Select **Stop if Applied**. The Rule Engine does not continue to the next rule in the security rule set if it matches the security rule and applies the rule aciton.

   Log When Rule is Applied

   Select the **Log When Rule is Applied** parameter to include log information in the Dynamic Data Masking system loggers when the Rule Engine applies the security rule.

   The following image shows the security rule parameters:



4. Click **OK** to close the Rule Editor.
5. Click **File** > **Update Rules** to save the rule and **File** > **Exit** to close the Rule Editor.

Step 6. View the Logs

View the log file to audit database access.

The appender saves the log file to the location that you specified in the appender **File** parameter. The log file contains information about the date, time, database, and log level of each security rule event. Each event includes the logger message that you defined in the security rule.

The following text shows example log output for the AccessLog.log file from this example:

```
10/15 15:18:04,056 [DDM for Oracle-2] INFO  - A database request occurred outside of business
hours.
10/15 15:18:04,060 [DDM for Oracle-2] INFO  - A database request occurred outside of business
hours.
```

```
10/15 15:18:04,063 [DDM for Oracle-2] INFO  - A database request occurred outside of business
hours.
10/15 15:18:04,067 [DDM for Oracle-2] INFO  - A database request occurred outside of business
hours.
```

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Acknowledgements

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